ExFEC: Evidence-Extended Factual Error Correction

情報学専攻　システム科学コース
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問題定義

- Factual Error Correction (FEC)：証拠に基づき文の誤りを特定し、最小限の修正で正しい情報に修正するタスク。

- **Article**: The Eiffel Tower is a wrought-iron lattice tower on the Champ de Mars in Paris.
- **Input Claim**: The Eiffel Tower is located in London.
- **Corrected Claim**: The Eiffel Tower is located in Paris.

関連研究

- **Mask-then-Correctアプローチ：**既存の手法はFEVERデータセットを活用し、「マスク・修正」アプローチに基づき、リモート監督で誤りを検出し、修正する。[1]
- 制限：修正の効果はマスクの効果に大きく依存する。[2]
- マスクの位置が誤っていると、修正できない。
- 一部の誤りは単語の置き換えだけでは直せない。
- データが足りない。

PivotFECによるFEI(Factual Error Injection)[2]

- Idea：FEIタスクを用いて多様な合成誤りのあるデータを生成し、FECモデルの学習を強化する。
- Pivotタスクの役割：FEIにより、少数の例でLLM（例：GPT）から誤りを含むテキストを生成し、データの量を向上。

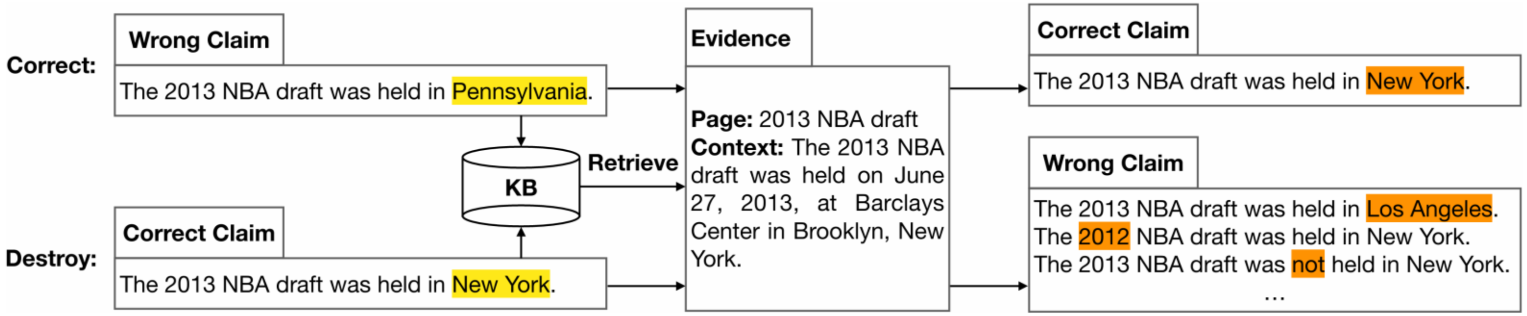
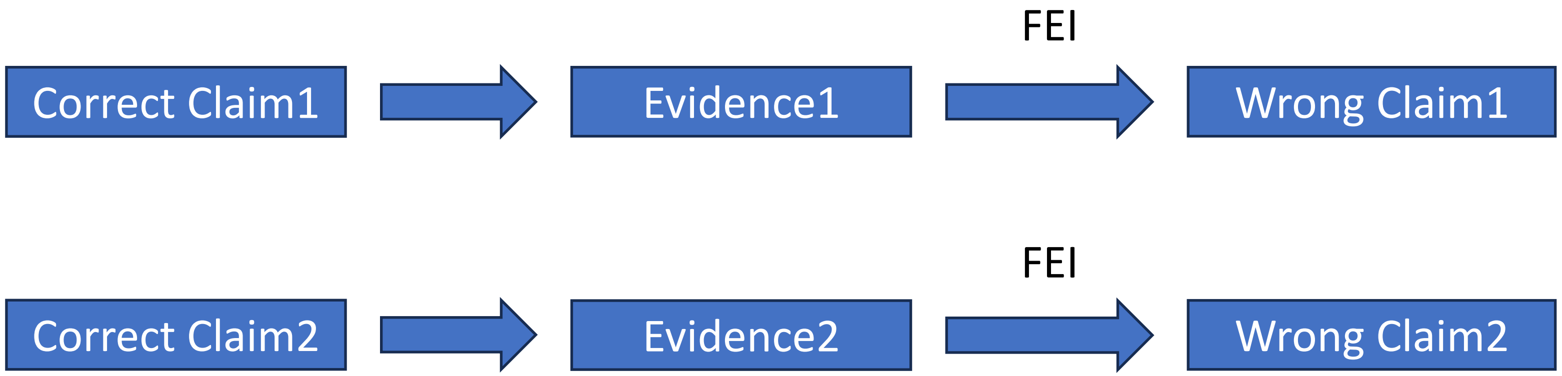


Figure 1: The comparison between factual error correction and factual error injection. From [2]

- 手順：
- 正確な主張を入力
- 誤り内容の挿入によるWrong(Mutated) Claimを生成

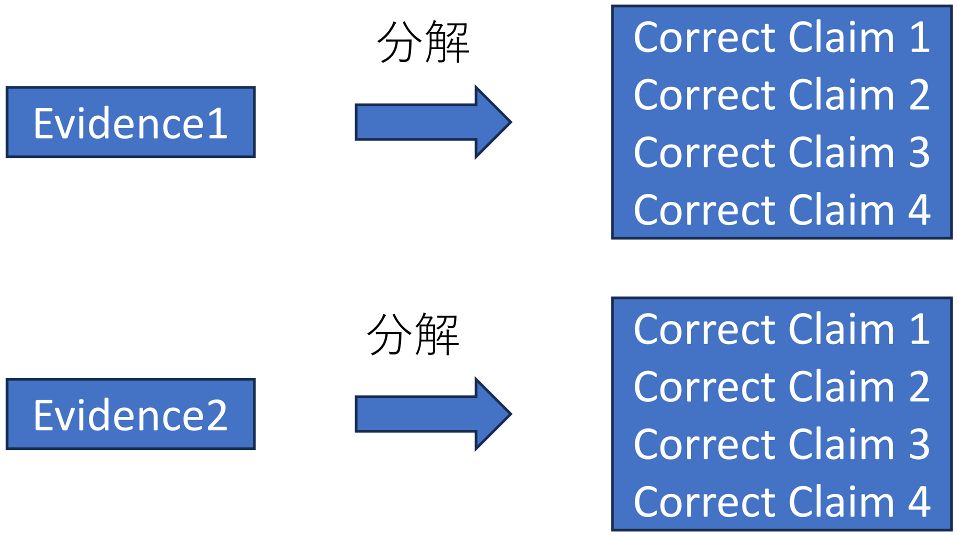
Proposed Step 1: Evidence Decomposition

- PivotFECの問題点：一つのCorrect claimから一つのWrong claimを生成する。しかし、Evidenceの数には限りがあるため、生成の効率はそれほど高くない。



Proposed Step 1: Evidence Decomposition

- 手法：セマンティクスや句読点の位置でEvidenceを分割し、複数のCorrect Claimに分解する



Evidence: One World Trade Center; One World Trade Center (also known as 1 World Trade Center , 1 WTC or Freedom Tower) is the main building of the rebuilt World Trade Center complex in Lower Manhattan , New York City .
World Trade Center (2001–present); The original World Trade Center featured the landmark Twin Towers , which opened in 1973 , and were the tallest buildings in the world at their completion .
Original claim: One World Trade Center opened in 2014.
Mutated claim: One World Trade Center opened in 1876.

PivotFEC From [2]

Evudence: “Machine learning’”, “Machine learning (ML) is a field of study in artificial intelligence concerned with the development and study of statistical algorithms that can learn from data and generalize to unseen data, and thus perform tasks without explicit instructions. Advances in the field of deep learning have allowed neural networks to surpass many previous approaches in performance.”
Split1: Machine learning (ML) is a field of study in artificial intelligence.
Split2: Machine learning focuses on the development and study of statistical algorithms that can learn from data.
Split3: In machine learning, statistical algorithms are designed to generalize to unseen data, enabling them to perform tasks without explicit instructions.
Split4: Advances in deep learning have significantly enhanced neural networks.
Split5: As a result, neural networks now outperform many previous approaches in terms of performance.

Proposed Approach: Evidence Decomposition

Desirable properties of decomposition.

- **一貫性**：分解後の各文の意味が元の証拠と同じであること。
- **完全性**：分解後の全ての文の内容を総合すると、元の証拠が表す内容と等しいこと。
- **独立性**：文脈がなくても、分解後の各文が意味のまとまりとして成立していること。

Evidence: “Qatar”, “Its sole land border is with Saudi Arabia to the south, with the rest of its territory surrounded by the Persian Gulf.”

Split1: Qatar’s sole land border is with Saudi Arabia to the south.

Split2: Besides Saudi Arabia, Qatar’s territory is surrounded by the Persian Gulf.

Proposed Step 2: Simplified FEI

- FEIの問題点： Evidenceは通常非常に長いため、Mutated Claimの生成に時間がかかる
- 提案：試行の結果、Evidenceを入力しなくてもLLMは高品質なMutated Claimを生成できることが判明した。そのため、Correct Claimのみを使用してMutated Claimを生成することとした。

Original claim1: The Lion King is about lions. Mutated claim1: The Lion King has nothing to do with lions.
Original claim2: The Lion King is a film. Mutated claim2: The Lion King is a TV show.
Original claim3: Carole King is an American. Mutated claim3: Carole King is an acrobat.
Original claim4: Bryan Adams has won Ivor Novello Awards. Mutated claim4: Bryan Adams has won Ivor Novello Awards for his singing.

Experiment

- Model: facebook/bart-base
- Early Stopping: Enabled, with patience set to 5 epochs.

Dataset	Total Instances	Training Instances	Validation Instances	Test Instances
PivotFEC	58,036	46,428	5,803	5,805
Our Dataset	9,786	7,828	978	980

Experiment

SARI: 文章の簡約化の品質を測る指標であり、追加（Add）、保持（Keep）、削除（Delete） の3つの操作に基づいて計算されます。(Higher is better.)

$$SARI = \frac{F1_{Add} + F1_{Keep} + F1_{Delete}}{3}$$

Dataset	SARI	Keep	Delete	Add	Eval Loss	Epochs
Mask then Correct†	0.447	0.652	0.559	0.128		
PivotFEC*	0.7782	0.7734	0.9880	0.5732	0.2333	16
Our Dataset*	**0.8632**	**0.9616**	**0.9960**	**0.6318**	**0.0692**	10

Results marked with † and ∗ are from Evidence-based Factual Error Correction and our reproduction.

Conclusion and Future Directions

Conclusion:

- Evidence Decompositionは、データ量を増やし、同じEvidenceに対する異なる視点のClaimを追加することで、FECタスクにおけるモデルの性能を大幅に向上させた。
- トレーニング方法だけでなく、データセットの最適化にも改善の余地がある
- 同じ証拠に対する繰り返し学習によって、モデルの文章理解が向上し、それに伴い性能も改善される。

引用

- [1]James Thorne and Andreas Vlachos. 2021. Evidence-based Factual Error Correction. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 3298–3309, Online. Association for Computational Linguistics.
- [2] PivotFEC: Enhancing Few-shot Factual Error Correction with a Pivot Task Approach using Large Language Models (He et al., Findings 2023)